FutureXcel Internship Program

Title:

Advanced Data Sales Dashboard Analysis Using Superstore Dataset

Role:

Data Science Intern

Organization:

FutureXcel

Week: 2

Data Visualization & Dashboard Creation

Prepared by:

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Tools & Technologies:

Python, Pandas, NumPy, Matplotlib, Seaborn, Jupyter Notebook

Submission Date:

12 February 2026

1. Introduction

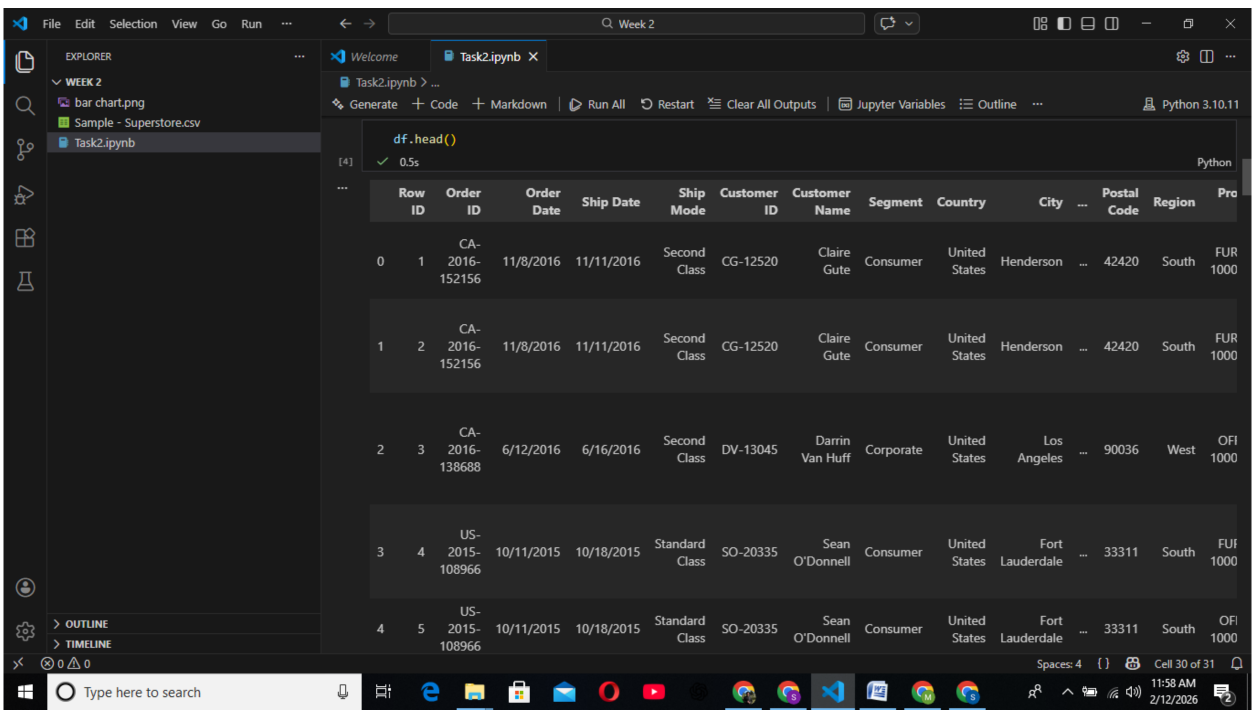
This report presents a dynamic visualization dashboard created using the Sample Superstore dataset. The purpose of this analysis is to understand business performance through multiple visualizations.

Different charts are used to answer important business questions related to sales trends, category performance, profit behavior, discount impact, and overall correlations.

The goal of this analysis is to:

- Analyze sales growth over time
- Compare category performance
- Study profit patterns
- Identify the impact of discounts
- Provide business recommendations

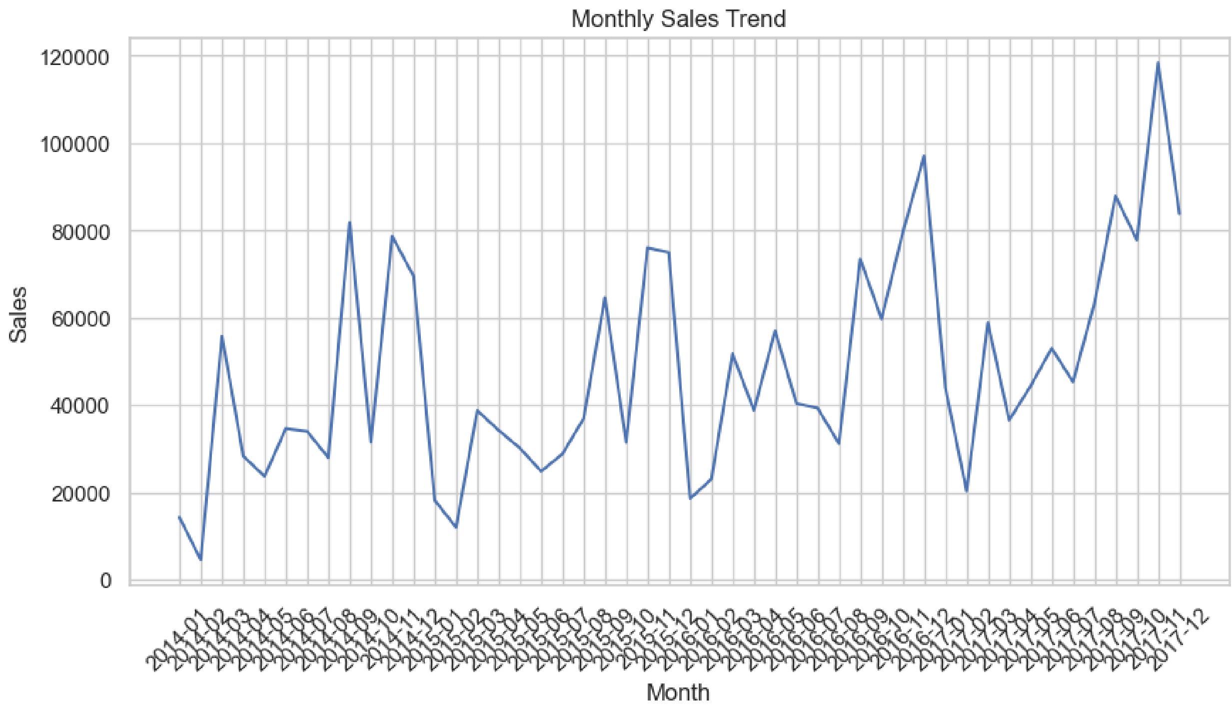
Dataset:



2. Monthly Sales Trend (Line Chart)

Business Question:

How do sales change over time?



Insights:

- Sales exhibit an overall upward trend over time.
- Several months show sharp spikes, indicating peak sales periods.
- Recurring fluctuations suggest a seasonal sales pattern.

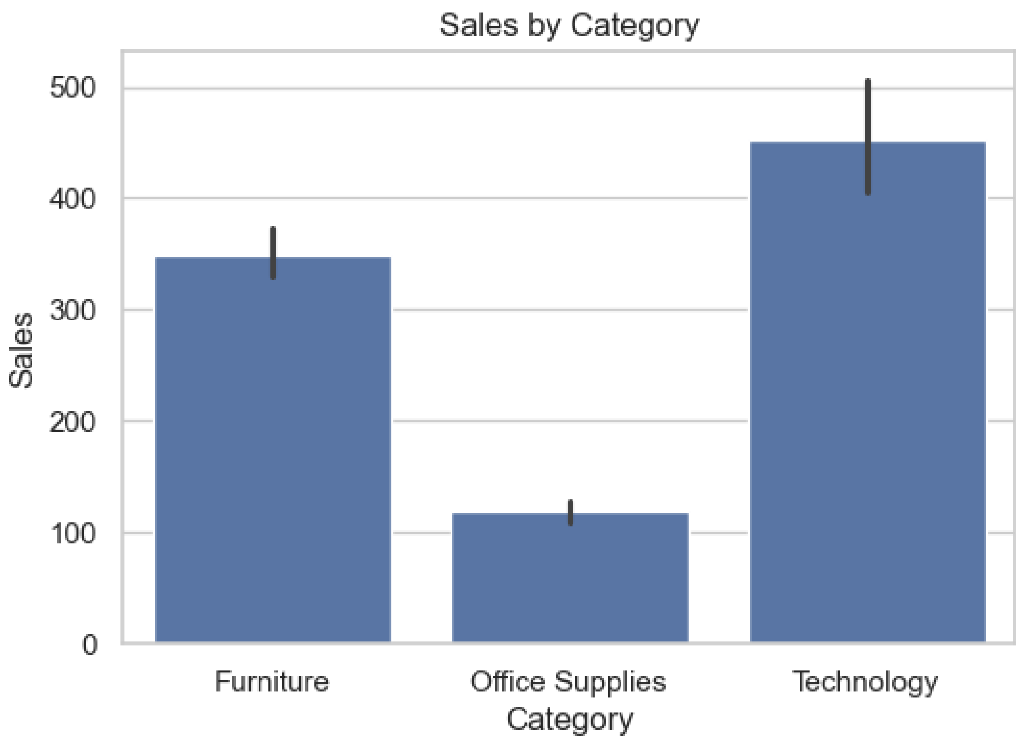
Interpretation:

The company shows steady growth, but seasonal patterns influence performance.

3. Sales by Category (Bar Chart)

Business Question:

Which category generates the highest sales?



Insights:

- Technology category generates the highest sales.
- Furniture is the second-highest sales category.
- Office Supplies has the lowest sales among all categories..

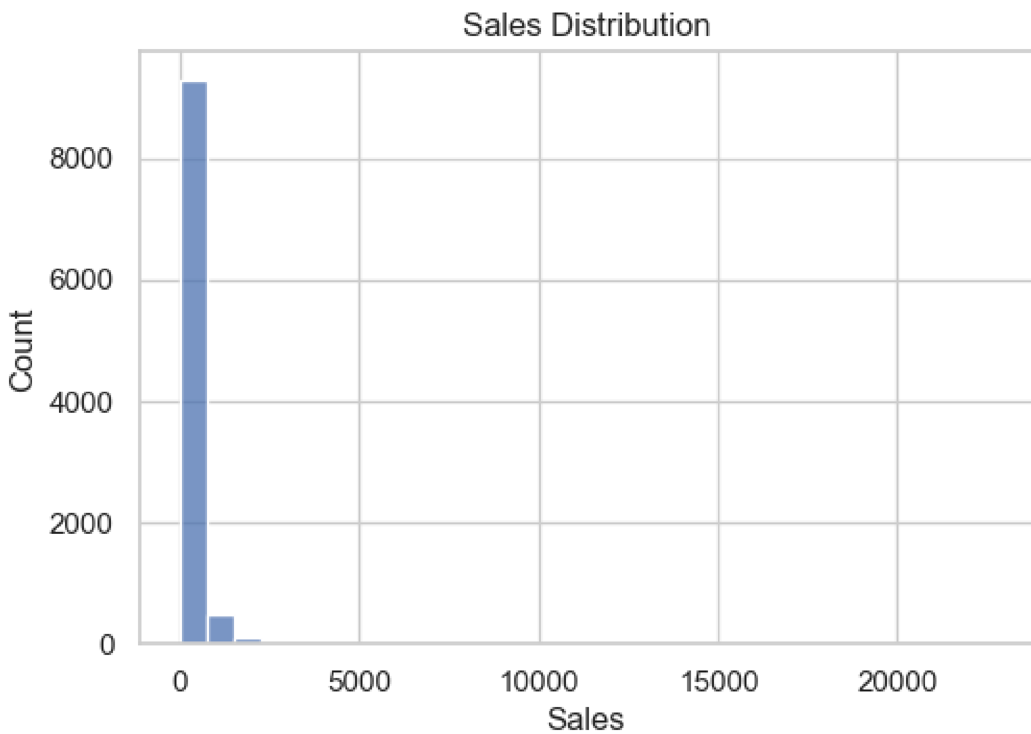
Interpretation:

Technology is the main revenue driver and should be prioritized.

4. Sales Distribution (Histogram)

Business Question:

How are sales values distributed?



Insights:

- Most transactions have low sales values.
- Few transactions have very high sales.
- The distribution is positively skewed.

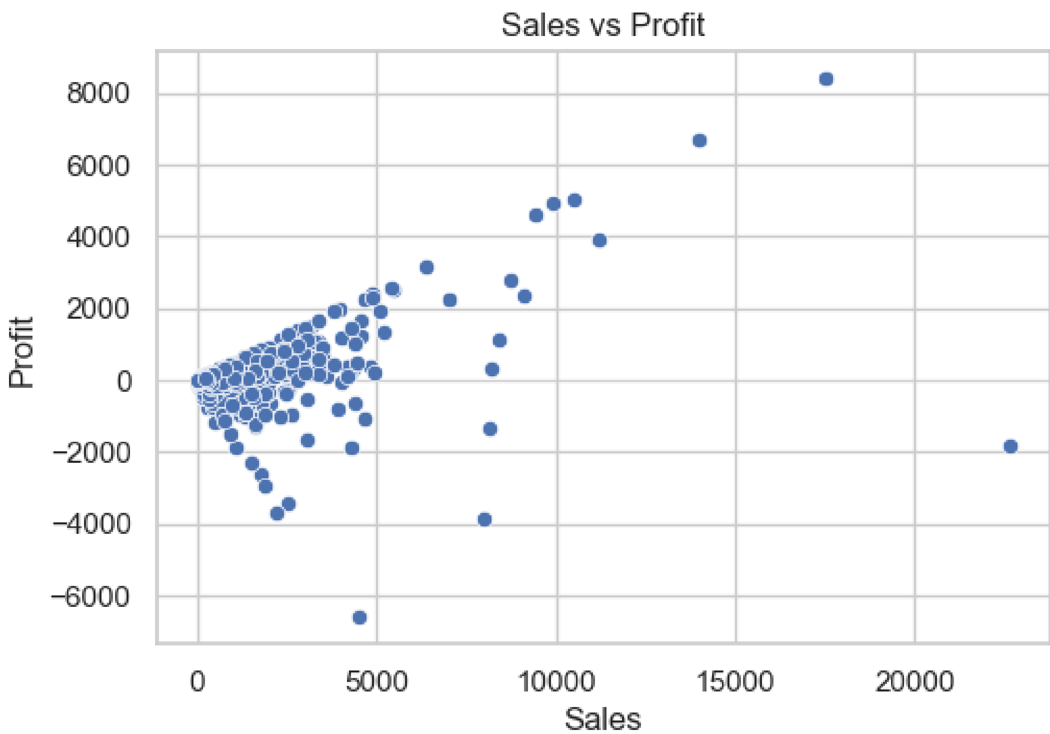
Interpretation:

Revenue depends heavily on a small number of high-value orders.

5. Sales vs Profit (Scatter Plot)

Business Question:

Is higher sales always equal to higher profit?



Insights:

- There is a positive relationship between sales and profit.
- Some high-sales orders generate losses.
- Profit does not increase perfectly with sales.

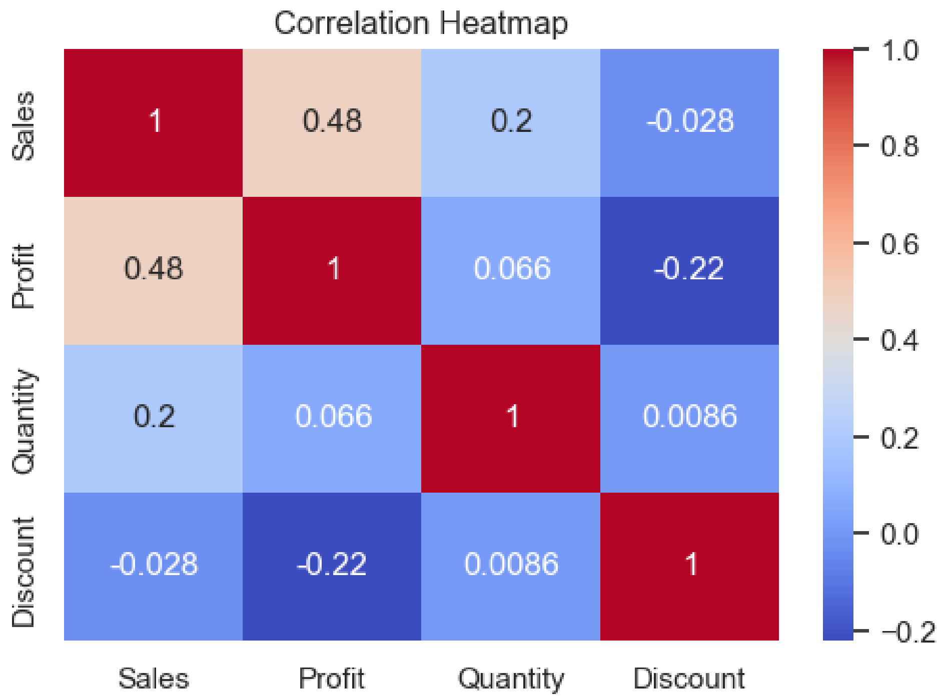
Interpretation:

Discount strategies need improvement to prevent losses.

6. Correlation Heatmap

Business Question:

What is the relationship between numerical variables?



Insights:

- Sales and Profit have strong positive correlation.
- Discount negatively impacts Profit.
- Quantity moderately affects Sales.

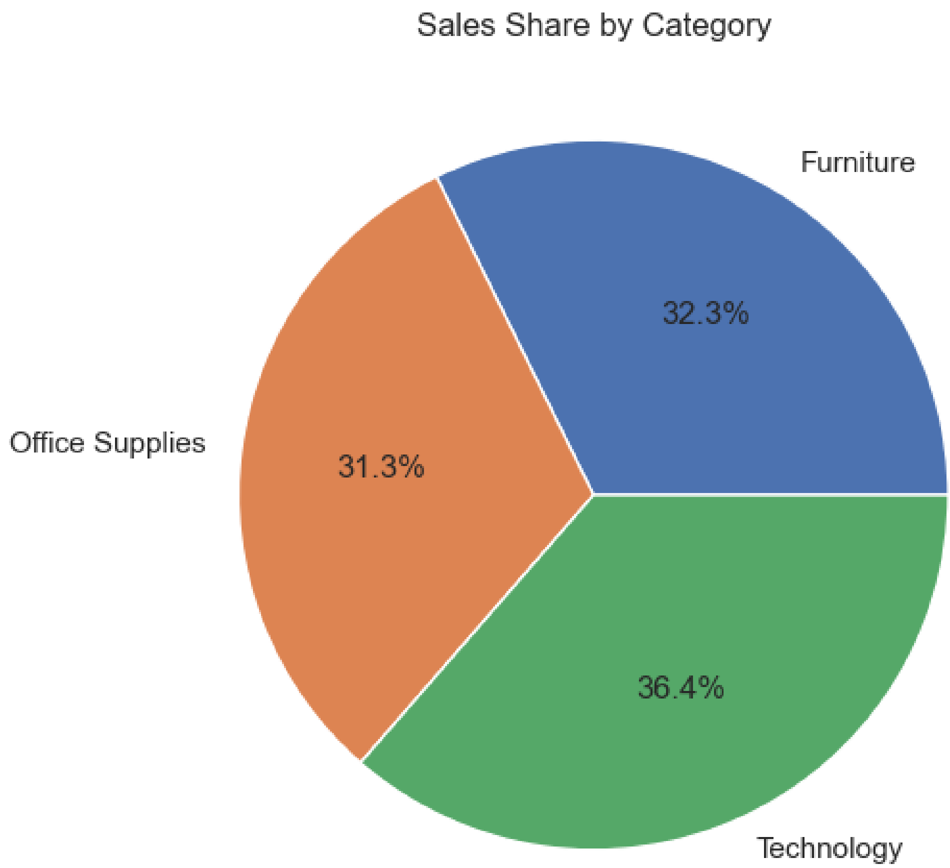
Interpretation:

High discounts reduce profitability and must be managed carefully

7. Sales Share by Category (Pie Chart)

Business Question:

What percentage of total sales comes from each category?



Insights:

• Technology holds the largest share of total sales (36.4%).
• Furniture is the second-largest category (32.3%).
• Office Supplies has the smallest share (31.3%).

Interpretation:

Technology dominates revenue contribution.

8. Final Conclusion

This dashboard analysis clearly shows that the company is growing over time with strong performance in the Technology category.

However, profitability is affected by high discounts in certain transactions. Seasonal trends are visible, and revenue is influenced by a small number of high-value orders.

Overall, the business is performing well but can improve profitability by optimizing discount strategies and focusing on high-performing product categories.